

Color merging

Chameleons can change their skin color rapidly to hide from prey and to signal to each other when they meet. They do this by expanding and contracting specialized color cells in the skin called chromatophores. The tiny chromatophores contain chemical pigments, and layers of several cells work together to reflect light and create flushes of different colors. The chameleon's nervous system controls these skin changes to mimic its environment or to show its mood.



Dressed up

Decorator crabs conceal themselves by attaching living animals to their bodies. This tropical species is called a soft coral crab because it collects pieces of coral to camouflage its shell. Tiny hooks on the crab's exoskeleton hold the corals firmly in place as they grow to form a living disguise. Unlike most camouflaged animals, decorator crabs lose their disguise when they molt.

To save time and energy, the crabs usually remove the camouflage from their old exoskeleton and carefully arrange it on their new one.



Crab's coloring blends in with its soft-coral covering

Insect disguises

More than nine-tenths of the world's insects spend their lives on plants, creating a huge food supply for insect-eating predators. The insects have evolved an incredible variety of disguises that help to make them almost invisible to hunters, even when seen up close. Large numbers of these plant-dwellers move and feed mostly after dark to keep from giving themselves away.



Leaf mimic

With its flattened body and mottled green markings, a leaf insect, or phyllid, looks almost identical to a real leaf. When it moves, it sways gently from side to side, just like a leaf rustling in the breeze.



Fake flower

In warm regions of the world, flower mantises lurk among colorful blooms, waiting for other insects to come their way. Their bodies are usually white or brightly marked, and they often have petal-like flaps on their legs.



Bark blender

Many moths seem to disappear the moment they land on bark. Their wings and bodies are colored like tree bark, and they may have contrasting stripes that break up their outline, making them even harder to see.



False thorns

Thorn bugs have a curved spike on their thoraxes, which looks exactly like a plant thorn. If a predator manages to see through this disguise, the spike makes the bugs very difficult to eat.



Living twig

The caterpillars of geometer moths look like tiny twigs covered with gray-brown bark. To complete the disguise, they rest with their bodies sloping outward, just like a real twig forking from a branch.